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Biogeochemistry of Natural Organic Matter

Global Meta-Analysis on the Relationship Between Mercury and Dissolved Organic Carbon in Freshwater Environments

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Key Points:

- Mercury and dissolved organic matter coupling is stronger in spatial studies, in North America, in natural systems, and in filtered samples
- Correlations are stronger with the aromatic fraction than the bulk dissolved organic carbon and stronger for total than methyl mercury
- Ecosystem type (lentic vs. lotic), geographical coordinates, and publication year had no effect on the strength of relationships

Supporting Information:

- Supporting Information S1
- Table S1

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(continued)

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Abstract In freshwater ecosystems, several studies have shown a strong linear relationship between total mercury (THg) or methylmercury (MeHg) and dissolved organic carbon (DOC) concentrations. Variations in this linear relationship have been reported, but the magnitude and causes of this variation are not well known. The objective of this study was to conduct a meta-analysis to quantify and understand the global variation of this mercury (Hg)–DOC association. This meta-analysis included 54 studies in lentic and lotic ecosystems for a total of 85 THg–DOC and 59 MeHg–DOC relationships. There was an increase in Hg with DOC concentrations in water with a global average slope of 0.25 (confidence interval (CI): 0.20–0.35) ng/mg for THg and 0.029 (CI: 0.014–0.044) ng/mg for MeHg. Relationships were stronger for (1) North American studies, (2) natural environments compared to those disturbed by anthropogenic activities, (3) spatial studies compared to temporal studies, (4) filtered samples (THg only), and (5) the aromatic fraction of DOC compared to the bulk DOC. Coupling with DOC was stronger for THg than for MeHg. Ecosystem type (lentic vs. lotic), geographical coordinates, and publication year did not influence the strength of relationships. Overall, we show that there is a strong but variable coupling between carbon and mercury cycles in freshwater ecosystems globally and that this link is modulated regionally by geographic location, temporal scale, and human activity, with implications for understanding these rapidly changing biogeochemical processes in response to global change.

Plain Language Summary In lakes and rivers, organic carbon is known to be a transporter of mercury, a toxic metal. However, depending on the chemistry of waterbodies, carbon can carry different amounts of mercury. This work compiled results of 54 scientific studies around the world looking at the correlation between mercury and organic carbon. We looked at the conditions that make this relationship vary. We found that relationships were almost always positive and that the type of carbon influenced the amount of mercury that was carried. The strength of those relationships was higher in natural ecosystems compared to those with human influence and in North American ecosystems compared to European and Asian ones. This work is important to understand the mechanism behind the association between mercury and carbon in different environments and how carbon can be used to explain variations in mercury, especially in a changing climate under human pressure.

1. Introduction

The cycles of mercury and carbon are strongly linked at the landscape and biogeochemical levels. In freshwater ecosystems, several studies have shown a strong linear relationship between total mercury (THg) or methylmercury (MeHg) concentrations and dissolved organic carbon (DOC) concentrations in water (Grigal, 2002; Riscassi & Scanlon, 2011; Stoken et al., 2016). This relationship is variable across and within systems, but the magnitude and causes of this variation remain poorly constrained, especially at the global scale, because of a lack of synthetic studies.

Several mechanisms have been documented for the mobilization and transformation of mercury (Hg) and DOC to, and in freshwater ecosystems, and many of them are shared. On one hand, Hg is a ubiquitous element with anthropogenic and natural sources (Pirrone et al., 2010). Contemporary anthropogenic emissions of Hg are mainly from artisanal and small-scale gold mining, coal burning, municipal incineration, and mining activities and have dramatically increased global atmospheric, lacustrine, and oceanic pools compared to the preindustrial era (Amos et al., 2015; Lamborg et al., 2014; Mason et al., 2012;

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Mason & Sheu, 2002). Once in aquatic environments, inorganic Hg can be methylated into methylmercury in anoxic conditions by a wide variety of microorganisms carrying the *hgcAB* gene cluster, such as sulfate-(SRB) and iron-(FeRB) reducing bacteria, methanogens and various types of *Firmicutes* (Compeau & Bartha, 1985; Gilmour et al., 1992; Gilmour et al., 2013; Parks et al., 2013; Podar et al., 2015). Once methylated, aqueous MeHg can bioaccumulate in organisms (Scheuhammer et al., 2007) and biomagnify in food webs (Lavoie et al., 2013). In freshwater ecosystems, mercury is transported from the catchment basin and can be methylated during its transit in wetlands, streams, rivers, and lakes. On the other hand, DOC is a complex and diverse group of carbon-containing compounds generally composed of humic substances (from the decomposition of plants and animals) as well as carbohydrates, carboxylic acids, amino acids, and hydrocarbons (Jaffé et al., 2008). DOC can have affinity with a wide range of trace metals including Hg (Mason, 2013). The origin (allochthonous vs. autochthonous) and composition (humic acid, fulvic acid, and humin) of DOC will impact the lability (labile vs. refractory), molecular weight, aromaticity, and amount of reduced sulfur binding sites (Bravo et al., 2017, 2018; Jiang et al., 2018; Lescord et al., 2018; Skyllberg et al., 2003). The propensity of Hg to bind with DOC will be determined by the amount of reduced sulfur available for binding (commonly measured by hydrophobic organic acid fraction; HPOA) and aromaticity (commonly measured by specific UV absorbance at 254 nm; SUVA₂₅₄ or by HPOA; Bravo et al., 2017; Burns et al., 2013; Dittman et al., 2009; Qian et al., 2002; Skyllberg et al., 2003; Xia et al., 1999). DOC structure will not only influence the binding and subsequent transport of Hg in watershed but also its bioavailability in aquatic organisms (Chiasson-Gould et al., 2014; French et al., 2014; Lescord et al., 2018; Tsui & Finlay, 2011). Hydrophobic fractions of DOC include humic and fulvic acids and have high reduced sulfur content (e.g., sulfides and thiols), which results in a preferred affinity with Hg (Qian et al., 2002; Skyllberg et al., 2003). Reduced sulfur binding sites contained in DOC exceed the amount of Hg available for binding in natural waters (Liem-Nguyen et al., 2017; Xia et al., 1999), thereby enhancing this association.

If Hg and DOC are coupled mechanistically, both elements should covary in natural ecosystems. Indeed, Hg–DOC relationships have been repeatedly observed across space (Braaten et al., 2014; Luo et al., 2014; MacMillan et al., 2015) and time (Burns et al., 2013; de Wit et al., 2014; Gerson & Driscoll, 2016) scales. MeHg–DOC relationships have been found to be weaker than THg–DOC relationships in the literature (Burns et al., 2012; Hall et al., 2008; Lescord et al., 2018), which could be due to the origin and molecular composition of DOC (Bravo et al., 2017, 2018) or to photodemethylation of MeHg in presence of DOC (Girard et al., 2016). Stronger relationships with Hg have been found for the aromatic fraction of DOC as measured by SUVA₂₅₄ or UV₂₅₄ (Burns et al., 2013; Riscassi & Scanlon, 2011) or by specific families of compounds measured by fluorescence and analyzed by parallel factor analysis (Jiang et al., 2018) than for bulk DOC concentrations. As a result, DOC is generally viewed as a vector of Hg at the catchment scale in freshwater ecosystems (Grigal, 2002; Ravichandran, 2004).

There is strong empirical support for a coupling between Hg and DOC, but variations in the Hg–DOC linear relationship among sites and among studies have been reported in the literature (Grigal, 2002; Riscassi & Scanlon, 2011; Stoken et al., 2016), and the causes are not well understood. Reviews have been published on Hg:DOC ratios and found a wide variation across the USA and Sweden (Grigal, 2002; Riscassi & Scanlon, 2011; Stoken et al., 2016). Stoken et al. (2016) found that the Hg:DOC ratio was strongly, negatively related to soil organic carbon (SOC) content, which may be explained by the fact that when carbon pools are in excess, there is a limited supply of Hg for adsorption relative to carbon (Stoken et al., 2016), which can be interpreted as a dilution of Hg by organic matter (OM). Although the role of DOC to act as a vector of Hg in freshwater systems at the landscape level is well known, it is unclear as to which factors control patterns over large spatial scales and whether anthropogenic activities modulate the magnitude of this relationship. Since OM is used for large-scale predictions of Hg concentrations in aquatic systems (Calder et al., 2016), a better understanding of our ability to accurately predict Hg based on DOC and the validity of such practice is needed. This is especially relevant with respect to anthropogenic engineering of waterways such as dams that disrupt the natural water flow and natural cycling of mercury. It is also important in the context of climate change that is expected to disturb the cycles of carbon and mercury (taken separately or in tandem) in freshwater ecosystems (Paranjape & Hall, 2017; Schuster et al., 2011).

The main objective of this study is to quantify and explain the coupling of Hg and DOC in freshwater ecosystems globally. In particular, the subobjectives are to (1) determine the strength (using coefficients of correlations) of the relationship between Hg and DOC, (2) determine the variables that explain the variations in the strength of relationships, (3) determine if the strength of relationships are stronger with the aromatic fraction of DOC compared to the bulk DOC, and (4) determine what explains variations in the Hg–DOC slopes and intercepts. The objectives 1–3 were accomplished using a conventional meta-analytical approach based on effect sizes (i.e., the coefficient of correlations (r) in this case). The objective 4 was based on equations (slopes and intercepts), which were put into various categories to explain variations.

2. Methods

2.1. Data Acquisition

We searched Web of Science™ for the keywords in the topic field TS = (*mercury and “dissolved organic carbon”) and a timespan from 2012 to 2017 (date of search 12 April 2017) to represent the last 5 years of scientific studies. The search resulted in 152 scientific articles, which were thoroughly read for potential inclusion in the database for the meta-analysis (see Figure S1 in the supporting information for decision of studies inclusion following Liberati et al., 2009 and Table S1 for studies included). Studies that were kept included results of the relationship between Hg concentrations (raw values) and DOC concentrations (raw values) in water. Freshwater systems (lakes, stream, reservoirs, and wetlands), which were impacted or not by anthropogenic activities (e.g., reservoirs, mining, urban area, and logging) or by natural phenomenon (e.g., beaver dams, and forest fires) were kept. Studies measuring relationships at several different locations and repeated measures on same locations were considered. Studies on marine, estuarine, and brackish systems were excluded, as were those on pore water, hyporheic water, soil, and sediments. Finally, controlled experiments where natural variation is not captured were excluded. Studies out of topic (e.g., when “mercury lamps” were used in a study) were also excluded. In total, 108 studies were excluded through that screening process. A total of 44 studies were kept from this search (Bergman & Bump, 2014; Berndt et al., 2016; Berndt & Bavin, 2012; Braaten et al., 2014; Bravo et al., 2017; Brazeau et al., 2013; Burns et al., 2012, 2013, 2014; Burton et al., 2013; Chaves-Ulloa et al., 2016; de Wit et al., 2014; Demers et al., 2013; Denkenberger et al., 2014; Dranguet et al., 2017; Eckley et al., 2015; Eklöf et al., 2013; Ekstrom et al., 2016; Fagnani et al., 2012; Gerson & Driscoll, 2016; Girard et al., 2016; Guedron et al., 2014; Hamelin et al., 2015; Haverstock et al., 2012; Houben et al., 2016; Jeremiason et al., 2016; Leopold et al., 2012; Loftin et al., 2012; Luo et al., 2014; MacMillan et al., 2015; Moreno et al., 2016; Navratil et al., 2015; Noh et al., 2016; Oswald & Branfireun, 2014; Painter et al., 2015; Picado & Bengtsson, 2012; Poste et al., 2015; Pyhtila et al., 2012; Riscassi et al., 2016; Sherman & Blum, 2013; Tjerngren et al., 2012; Ukonmaanaho et al., 2016; Vidon et al., 2014; Voss et al., 2015). We included additional studies that have been cited by the abovementioned pool of studies more than five times specifically in reference to the relationship (or lack thereof) between Hg and DOC (eight studies; Brigham et al., 2009; Dittman & Driscoll, 2009; Dittman et al., 2009; Dittman et al., 2010; Driscoll et al., 1995; Eklöf et al., 2012; Porvari & Verta, 2003; Riscassi & Scanlon, 2011; Sørensen et al., 2009). Studies cited for their absence of relationship were specifically included in the meta-analysis to avoid publication bias (also referred to as the “file drawer effect”), which consist of omitting studies that did not report significant results (Borenstein et al., 2009). Studies included in this meta-analysis were summarized in Table S1 and plotted in Figure 1. Average coordinates were plotted for studies conducted on multiple sites.

2.2. Data Treatment and Statistical Analyses

When equations were not directly available in studies, two methods were used to generate equations. First, raw data were sometimes available in tables or in supporting information and were used to calculate equation parameters. Second, data were extracted from biplot figures (Hg vs. DOC) using *Datagrabber* (Roxburgh, 2006) or *WebPlotDigitizer* (Rohatgi, 2017). When equations were reported in a less common unit (e.g., moles) or using transformations (e.g., logarithm), data extraction software packages were also used to convert into grams and raw data, respectively. Details on data extraction accuracy and precision are found in supporting information (Text S1).

2.2.1. Strength of the Relationship and Factors

The *metaphor* package for meta-analysis in *R* (<http://www.metafor-project.org/doku.php>; Viechtbauer, 2010) was used to test the strength of relationships (objective 1) and determine which variables controlled

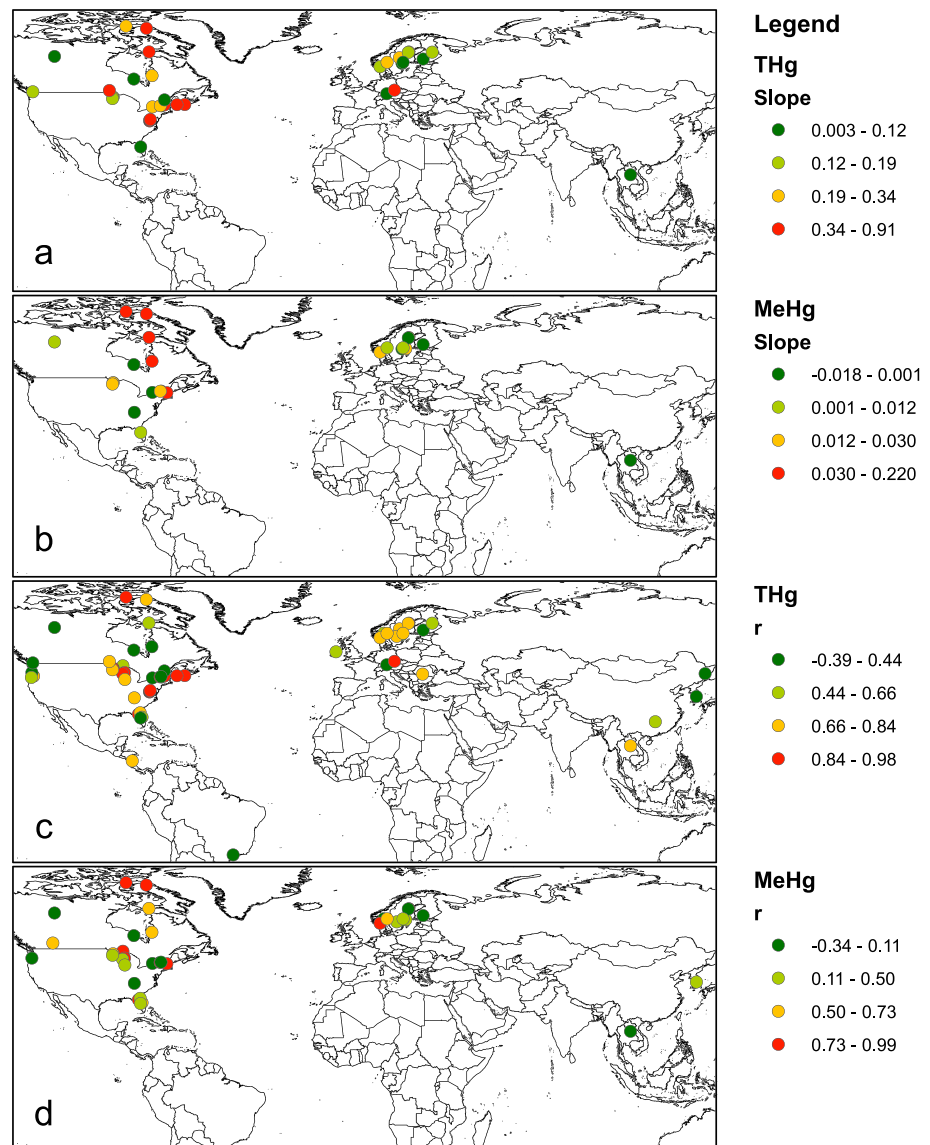


Figure 1. Location of studies included in the meta-analysis (a) THg slopes, (b) MeHg slopes, (c) THg coefficient of correlation (r), and (d) MeHg r . Color of points represents the strength or slopes of Hg–DOC relationships. Note that some studies are conducted over broad areas; in these cases, average coordinates are provided.

variations in the association Hg–DOC (objective 2). In meta-analyses based on regressions, statistics are performed on coefficients of correlation (r ; Borenstein et al., 2009)). Coefficients of correlations can be used for comparisons of relationships based on different ranges (e.g., Hg–DOC vs. Hg–SUVA₂₅₄). Correlation coefficients were transformed into the Fisher's z scale for statistical tests and figures:

$$z = 0.5 \times \ln \left(\frac{1+r}{1-r} \right) \quad (1)$$

and backtransformed to r for discussion (Borenstein et al., 2009):

$$r = \frac{e^{(2 \times z)} - 1}{e^{(2 \times z)} + 1} \quad (2)$$

Random effect models were run for overall estimate of correlations with the z statistics. Differences between studies (e.g., due to differences in methods) are expected in meta-analyses and can introduce variability,

referred to as “heterogeneity” which was measured by the Cochran’s χ^2 test (Q test; Higgins & Thompson, 2002; Higgins et al., 2003; Viechtbauer, 2010). A measure of variance (or “inconsistency”) was also reported (I^2) to describe the proportion of total variation across studies that is due to heterogeneity rather than chance (Higgins et al., 2003; Higgins & Thompson, 2002) and I^2 can also be interpreted as the proportion of observed variation that can be attributed to the actual difference between studies, rather than within-study variance (Quintana, 2015). Higgins et al. (2003) suggested that I^2 values of 25%, 50%, and 75% are low, moderate, and high within-study variance, respectively. Mixed effect models were used to test hypotheses by including variables, called “moderators” or “study level variables” (continuous or categorical) in meta-analyses, to determine if Hg–DOC relationships depend on the characteristics of the studies (Viechtbauer, 2010). Moderators were included in the model to account for at least part of the heterogeneity in the true effects and tested using the omnibus test (Q_M) against a χ^2 distribution. When moderators were used in the model, the test of residual heterogeneity (Q_E) was used to determine if other moderators not considered in the model could be influencing the Hg–DOC relationship and are presented in supporting information (Text S2).

To evaluate the influence of publication bias, the “trim and fill” method of the metaphor package (Viechtbauer, 2010) was used, which provides an estimate of the number of studies that are missing from the meta-analysis and adjusts the overall effect size mean if needed. It should not be interpreted as a more valid estimate of the overall effect but rather to examine the sensitivity of the results to the publication bias. In addition, the Egger regression test (Egger et al., 1997) was carried out to determine asymmetry (Viechtbauer, 2010).

2.2.2. Comparison of Aromatic and Bulk DOC

For the difference between Hg–DOC and Hg–Aromatic portion of DOC (UV_{254} , $SUVA_{254}$, UV_{420} , and UV_{320}) in objective 3, we ran a multivariate model with *DOC type* as random (repeated) variable nested within the *Study* using the function *rma.mv* in the *metafor* package (Viechtbauer, 2010). We used studies that measured both DOC and a measure of aromaticity, which led to 10 studies with 16 pairwise comparisons for THg (Bravo et al., 2017; Burns et al., 2013; Dittman et al., 2009; Eklöf et al., 2012; Eklöf et al., 2013; Jeremiason et al., 2016; MacMillan et al., 2015; Navratil et al., 2015; Riscassi et al., 2016; Riscassi & Scanlon, 2011) and 4 studies with 6 pairwise comparisons for MeHg (Bravo et al., 2017; Eklöf et al., 2013; Jeremiason et al., 2016; MacMillan et al., 2015). The most widespread measure of aromaticity was $SUVA_{254}$, which is the specific UV absorbance at 254 nm divided by DOC concentration (UV_{254}/DOC). We also included other UV absorbance measures such as 254 (UV_{254}), 320 (UV_{320}), and 420 nm (UV_{420}).

2.2.3. Comparing Relationships Across Treatments and System Types

In order to test whether certain conditions alter the relationship between Hg and DOC, we defined categories related to geographic location, system type, and sampling methods to include as predictors in the meta-analysis (see Figure 2 and Tables 1 and S1 for levels of variables). “Continent” included (1) Asia, (2) Europe, (3) North America, and (4) South America. “Ecosystem type” referred to whether samples were collected in (1) lentic or (2) lotic ecosystems. “Sampling design” referred to whether samples were collected (1) once for several sites (spatial), (2) several times at the same site (temporal), or (3) several times at several sites (spatiotemporal). “Human disturbance” was divided into categories that were believed to differ in their Hg or DOC input in the systems: (1) Natural (possibly low Hg and DOC), (2) Logging (high DOC and possibly high Hg), (3) Mine/Coal/Chlor-alkali industries (high Hg but no reason to believe that DOC would be high), and (4) Urban/Industries (possibly high DOC and Hg). For the latter category, we could not easily separate urban from industries because industries were often in urban areas. Finally, “Filtration” refers to whether Hg samples were (1) filtered (0.2 to 0.7 μ m pores depending on studies) or (2) not filtered.

2.2.4. Variations in Slopes and Intercepts

High slopes (b) and intercepts (a) in relationships between Hg and DOC, respectively, indicate (1) high mercury concentrations per unit of DOC and (2) the theoretical concentration of Hg in the absence of DOC. However, intercepts are mathematically negatively related to slopes (Froese, 2006; Lavoie et al., 2013; Prairie et al., 1995) and should therefore be interpreted with caution. Moreover, negative intercepts have no practical or logical meaning in the case of Hg–DOC relationships.

Models were run with *slope* or *intercept* as the dependent variables. Each variable that was a significant predictor of the strength of relationships (based on outcome of objective 2) was used as a variable to explain variations of the parameters. Model selection was done by Akaike Information Criterion corrected (AICc) for small sample sizes to determine the most parsimonious model using the Multi-Model Inference (MuMIn)

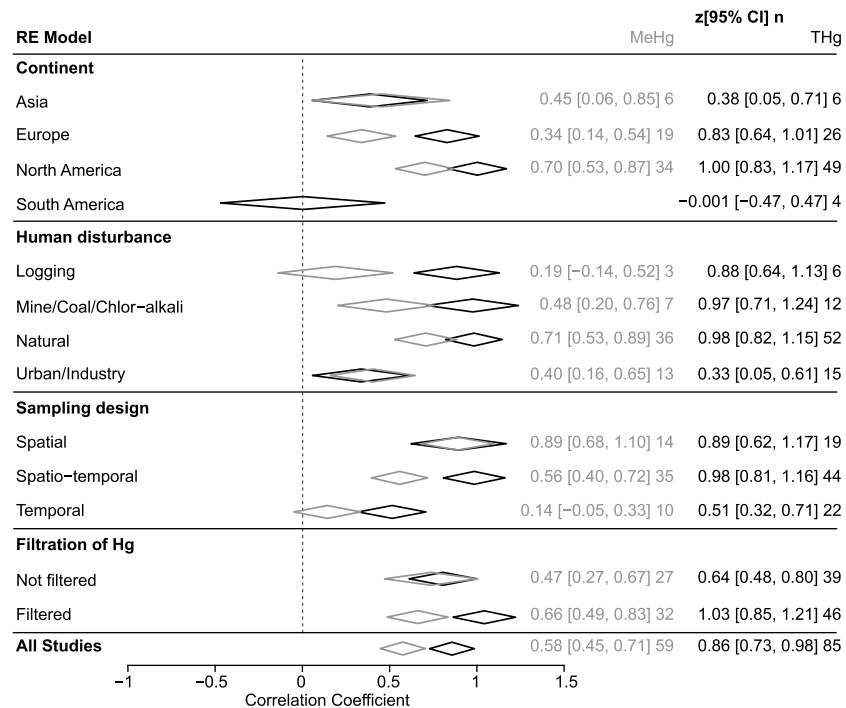


Figure 2. Summary forest plot of z scale correlation coefficients (z [95% confidence intervals (CI)] n) of MeHg–DOC relationships (grey) and THg–DOC relationships (black) for each level of moderator variables of interest taken separately in random models.

package (Barton, 2013) based on a set of candidate models of main factors. Since the difference in AICc (Δ AICc) between the best model and the second-best model was low (<4) and the Akaike weight (w ; model probability) of the top model was low, we calculated the model average based on the upper 0.95 cumulative sum of w . We also calculated the relative variable importance, which is the number of models with a specific variable of interest in proportion to all possible models included in the upper 0.95 w . To determine the empirical support of a model over another one, we calculated evidence ratios, which is the w of top model over the w of another model such as the second-best model or last model (Burnham et al., 2011).

3. Results

3.1. Results of the Search

Data were extracted from 52 published papers: 48 including THg–DOC relationships and 31 including MeHg–DOC relationships. We extracted 85 effect size (r) values for THg and 59 for MeHg. These effect sizes came from 14 countries on four continents with a strong bias for North American studies (58% for both THg and MeHg). A total of 50 THg and 40 MeHg slopes and intercepts were extracted.

3.2. Strength of the Hg–DOC Relationship

This meta-analysis showed a strong evidence of linear (rather than logarithmic or exponential) Hg–DOC relationships. There was a significant positive relationship, strong in magnitude (Cohen, 1992), between THg and DOC ($z = 13.1$, $p = <0.001$; $r = 0.694$ [0.621–0.755] calculated from Fisher's z scale: 0.856 [0.727–0.984]). Relationships between MeHg and DOC were significant and positive ($z = 8.7$, $p = <0.001$) but weaker than for THg ($r = 0.520$ [0.420–0.608]). There was a high heterogeneity for THg ($\tau^2 = 0.300$, $Q = 1,161$, $df = 84$, $p < 0.001$) indicating significant between-study variability in r and high within-study variance (I^2 : 94% [92%–96%]), which shows that a large proportion of variation remained unexplained. The funnel plot indicated a good distribution of studies and the Egger regression test confirmed that there was no asymmetry ($z = -0.30$, $p = 0.77$) suggesting that there was no publication bias in this meta-analysis. For

Table 1
Median and Mean ($\pm$ SD and $\pm$ 95% CI) for Slopes and Intercepts

	Slope				Intercept				N studies	N samples
	Median	Mean	SD	CI	Median	Mean	SD	CI		
THg										
Overall	0.19	0.25	0.20	0.06	0.52	0.59	1.35	0.37	50	3578
Continent ^a										
Asia	0.12	0.12	0.01	0.01	−0.11	−0.11	0.40	0.55	2	15
Europe	0.14	0.18	0.21	0.10	1.59	1.25	1.62	0.77	17	1590
North America	0.25	0.30	0.19	0.07	0.26	0.28	1.08	0.38	31	1973
Human disturbance										
Logging	0.15	0.15	0.00	0.01	1.88	1.88	0.08	0.11	2	218
Mine/Coal/Chlor-alkali	0.19	0.26	0.29	0.22	−0.15	−0.14	0.98	0.72	7	313
Natural	0.25	0.28	0.20	0.06	0.73	0.70	1.47	0.49	35	2503
Urban/Industry	0.12	0.12	0.08	0.06	0.29	0.42	0.66	0.53	6	544
Sampling design										
Spatial	0.26	0.24	0.17	0.09	0.52	0.44	0.94	0.49	14	293
Spatiotemporal	0.21	0.29	0.23	0.08	0.17	0.42	1.33	0.49	28	3173
Temporal	0.14	0.12	0.06	0.04	1.94	1.47	1.77	1.23	8	112
Hg filtered										
Not filtered	0.16	0.18	0.15	0.06	0.85	1.10	1.37	0.56	23	1967
Filtered	0.25	0.31	0.23	0.08	−0.04	0.16	1.19	0.45	27	1611
MeHg										
Overall	0.012	0.029	0.050	0.015	0.013	−0.019	0.584	0.181	40	1568
Continent ^a										
Asia	0.005	0.004	0.007	0.007	0.013	−0.022	0.106	0.104	4	29
Europe	0.009	0.015	0.033	0.016	0.204	0.156	0.744	0.365	16	471
North America	0.023	0.045	0.060	0.026	−0.018	−0.158	0.464	0.203	20	1068
Human disturbance										
Logging	0.001	0.001			0.323	0.323			1	187
Mine/Coal/Chlor-alkali	0.012	0.009	0.007	0.005	0.034	0.075	0.112	0.083	7	421
Natural	0.021	0.042	0.060	0.024	0.001	−0.097	0.733	0.293	24	508
Urban/Industry	0.004	0.011	0.021	0.015	0.042	0.091	0.259	0.179	8	452
Sampling design										
Spatial	0.026	0.058	0.074	0.042	−0.030	−0.239	0.616	0.349	12	281
Spatiotemporal	0.011	0.015	0.019	0.008	0.013	0.070	0.195	0.085	20	1209
Temporal	0.008	0.021	0.046	0.032	0.221	0.088	1.041	0.721	8	78

Note. CI = confidence interval; SD = standard deviation.

^aSlopes and intercepts were not available for South America.

MeHg, there was also a high heterogeneity ($\tau^2 = 0.192$, $Q = 474$, $df = 58$, $p < 0.001$) and high variance (89% [83%–93%]). As for THg, MeHg revealed a good distribution on the funnel plot and no asymmetry ($z = 0.66$, $p = 0.51$). Negative relationships were found for only 8% (7/85) of all cases for THg (Brigham et al., 2009; Fagnani et al., 2012; Noh et al., 2016; Porvari & Verta, 2003; Vidon et al., 2014) and 12% (7/59) for MeHg (Bravo et al., 2017; Brazeau et al., 2013; Guedron et al., 2014; Porvari & Verta, 2003; Riscassi et al., 2016).

3.3. Controls on the Hg–DOC Relationships

A combination of geographic and treatment variables was important in explaining the differences in Hg versus DOC relationships. In particular, the variables *continent*, *human disturbance*, *filtration* (for THg only), and *sampling design* (see Figure 2 and Table 1 for levels of variables) were significant, whereas *latitude* and *longitude*, *ecosystem type* (lentic ($n = 23$ for THg and 17 for MeHg) versus lotic ($n = 62$ for THg and 42 for MeHg)), and *publication year* were not significant for both THg and MeHg. When significant variables (continent, human disturbance, filtration (for THg only), and sampling design) were included in a common model, it was significant ($Q_M = 44.6$, $df = 9$, $p < 0.001$) and the percent of the total amount of heterogeneity that can be accounted for by including those moderators was $R^2 = 36.2\%$ (i.e., about 40% of the variation in the coefficient of correlations (r) could be accounted for by these four predictors). In the case of MeHg, moderator variables were significant ($Q_M = 22.9$, $df = 5$, $p < 0.001$) and the R^2 was 28.8%. Tests of residual

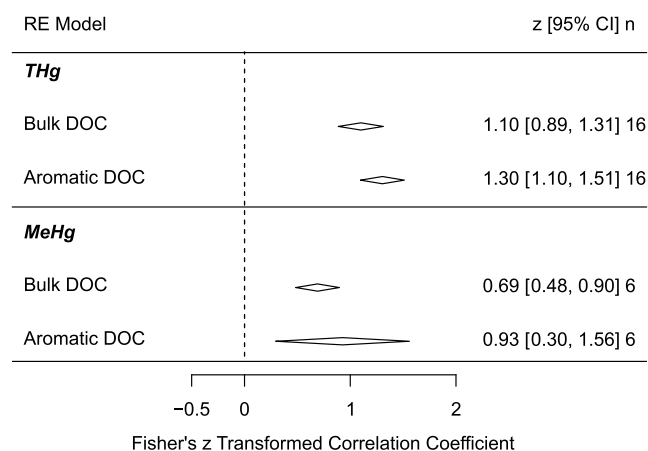


Figure 3. Summary forest plot of z -scale correlation coefficients (z [95% confidence intervals (CI)] n) for relationships between Hg (THg and MeHg) and the aromatic fraction of DOC or the bulk DOC. CI = confidence interval; DOC = dissolved organic carbon.

heterogeneity (Text S2) indicated that other moderators not considered in the model could be influencing THg–DOC and MeHg–DOC relationships.

Each level of moderator variables of interest was taken separately in random effect models (Figure 2). Overall, THg and MeHg showed similar patterns among levels of the moderator variables: Coefficients of correlation were high for North America studies, when relationships were based on studies on a spatial gradient (rather than on a temporal or a spatiotemporal gradient) and in natural setting (i.e., with no direct human influence). For the few studies conducted in South America, relationships were not significant. There was a high level of heterogeneity ($Q > 17$; all $p < 0.01$) and variance for THg (I^2 from 74.6% to 96.6%) with only Asia being below the 75% mark for moderate variance. For MeHg, there was less heterogeneity and lower I^2 values (25.8% to 93.4%) with two cases where variables were not significant and with moderate variance: studies conducted in Asia ($Q = 10.8$, $df = 5$, $p = 0.06$; $I^2 = 54.3\%$) and studies conducted on a temporal scale ($Q = 12.2$, $df = 9$, $p = 0.20$, $I^2 = 25.8\%$), showing that there was lower within-study variance for MeHg in those cases. With the different tests used, relationships between MeHg and DOC were generally weaker (lower r) than those between THg and DOC.

3.4. Comparison of Aromatic and Bulk DOC

The relationship between Hg and optical indices of DOC related to aromaticity (e.g., $SUVA_{254}$) was reported in a subset of studies; hence, it was possible to compare the relationships between Hg and total DOC to the relationships between Hg and aromatic DOC. We found that the aromatic fraction of DOC (i.e., $SUVA_{254}$, UV_{254} , UV_{320} , and UV_{420}) was more strongly related to THg than bulk DOC ($Q_M = 7.3$, $df = 1$, $p = 0.007$; Figure 3). Average (95% CI) coefficients of correlation (r) for relationships with the aromatic fraction were 0.842 (0.789–0.882), while relationships with bulk DOC were 0.817 (0.758–0.862). MeHg was also more strongly correlated with aromatic DOC ($Q_M = 18$, $df = 1$, $p < 0.001$). Aromatic fraction had an r average of 0.745 (0.523–0.872), whereas for bulk DOC, it was 0.600 (0.305–0.790). There was no publication bias in this component of the meta-analysis (Egger regression test: THg: $z = -0.83$, $p = 0.41$; MeHg: $z = -1.6$, $p = 0.11$). Test of residual heterogeneity indicated that other moderators could explain variations for THg and MeHg relationships (Text S2).

3.5. Variations in Slopes and Intercepts

Slopes and intercepts were highly variable (Figures 4 and 5 and Table 1). The variables that were significant in section 3.3 were used as explanatory variables, and slopes (b) or intercepts (a) were used as dependent variables in linear models to explain variations. When continent, human disturbance, sampling design, and filtration were taken together, the full model explained 33% of the variation in THg slope (linear model: $F_{(8,41)} = 2.6$, $p = 0.023$). The Akaike weight (w ; i.e., model probability) of the best model based on lowest AIC was only of 0.311, forcing us to use model averaging. The model average based on the 0.95 cumulative sum of w showed that variability was explained in order of importance by the filtration (79% of models contained this variable), sampling design (56%), human disturbance (53%), and continent (18%). The evidence ratio showed that the best model was 4.9 times more likely than the second-best model and 139 times more likely than the worst model (out of 16 models). For THg intercepts, the full model was not significant ($F_{(8,41)} = 1.5$, $p = 0.202$). The same procedure as for the slope was nevertheless used. The AIC weight was 0.408 for the top model and model averaging showed that variability was explained by the filtration (72%), continent (32%), sampling design (22%), and human disturbance (8%). The evidence ratio showed that the best model was 2.9 times more likely than the second-best model and 740 times more likely than the worst model.

For MeHg slopes, the full model explained 38% of the variation (linear model: $F_{(7,32)} = 2.8$, $p = 0.020$). The w of the best model was only 0.319 and the evidence ratio showed that the best model was 4.5 times more likely than the second-best model and 23 times more likely than the worst model. Model averaging showed that variability was explained by continent (66%), sampling design (49%), and human disturbance (42%). For MeHg intercepts, the full model was not significant ($F_{(7,32)} = 1.1$, $p = 0.406$). The AIC weight was 0.523

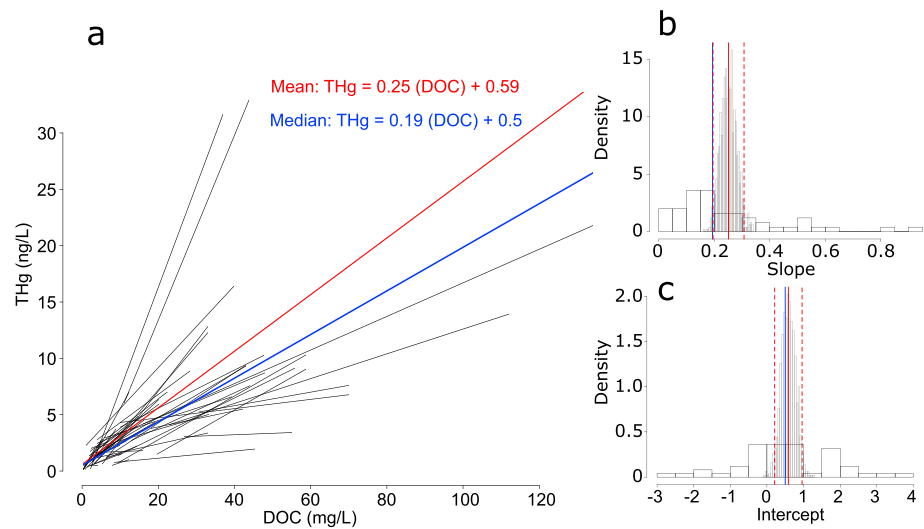


Figure 4. (a) Relationships between total Hg (THg) and DOC. The mean (red line) and median (blue line) slopes and intercepts are shown. Histograms of the distribution of (b) slopes and (c) intercepts of studies (large black boxes) and bootstrap of mean ($N = 1,000$; thin grey boxes). Mean bootstrap (red line) $\pm 95\%$ confidence intervals (dashed red lines) and median bootstrap (blue line) are shown. DOC = dissolved organic carbon.

for the top model and the evidence ratio showed that the best model was 2.8 times more likely than the second-best model and 281 times more likely than the worst model. Model averaging showed that variability was explained by continent (28%), sampling design (24%), and human disturbance (6%).

4. Discussion

This meta-analysis shows that Hg (total and MeHg) and DOC are correlated in freshwaters globally and that there are consistent patterns in these relationships depending on geographic features, temporal scales, and sampling treatments (e.g., filtration). The global estimate of THg per unit DOC was 0.25 ng/mg, and the

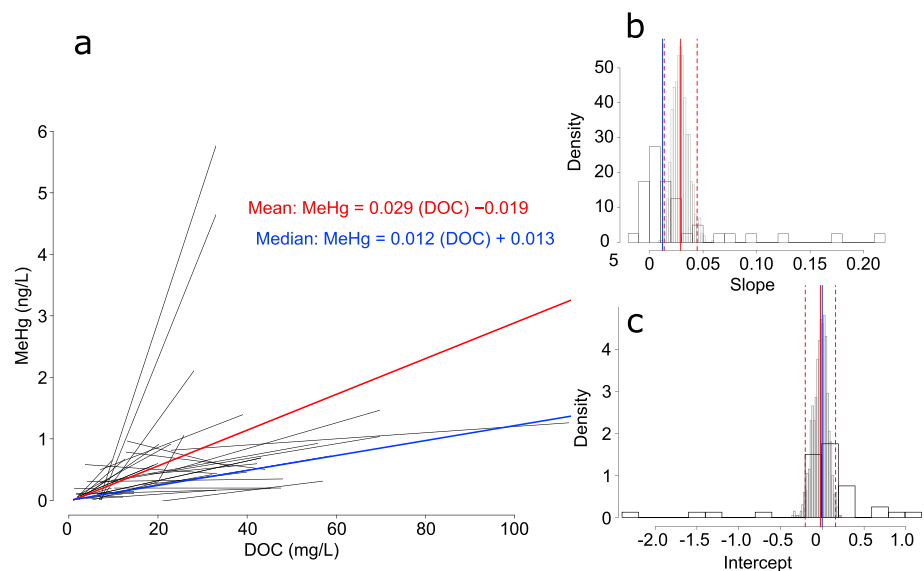


Figure 5. (a) Relationships between methylmercury (MeHg) and DOC. The mean (red line) and median (blue line) slopes and intercepts are shown. Histograms of the distribution of (b) slopes and (c) intercepts of studies (large black boxes) and bootstrap of mean ($N = 1,000$; thin grey boxes). Mean bootstrap (red line) $\pm 95\%$ confidence intervals (dashed red lines) and median bootstrap (blue line) are shown. DOC = dissolved organic carbon.

global estimate of MeHg per unit DOC was 0.029 ng/mg. There was a stronger association between both forms of mercury with aromatic DOC than with total bulk DOC.

The variable coupling of both elements is consistent with previous studies that have documented mechanisms responsible for chemical and hydrological coupling in natural environments. For example, the binding of Hg to DOC is mainly controlled by the quality and type of OM (Lescord et al., 2018; Ravichandran, 2004). Hg has a high affinity for molecules containing reduced sulfur such as thiols, and their availability in the system may control Hg–DOC associations and cotransport (Haitzer et al., 2002; Ravichandran, 2004; Skyllberg et al., 2003). Strong relationships have been associated with similar sources of Hg and DOC (e.g., from soil or wetland) resulting in a cotransport, whereas weak relationships have been associated with external sources of Hg (e.g., atmospheric deposition or point sources) or when DOC binding sites are limited (Ravichandran, 2004). In some cases (e.g., in beaver dams), lack of Hg–DOC relationships would indicate that Hg dynamics are modulated by reducing conditions rather than by DOC patterns (Roy et al., 2009). Although wetlands are a source of MeHg and DOC for downstream ecosystems (Burns et al., 2012; Wiener et al., 2006), the cotransport with DOC in those systems is not as clear. Good correlations between Hg_D or $MeHg_D$ and DOC have been found in many wetlands (Demers et al., 2013; Guedron et al., 2014; Hurley et al., 1995, 1998), but a correlation was not found in the Everglades wetlands ($r < 0.01$), despite a wide range in DOC concentrations (20–60 mg/L) (Hurley et al., 1998). The Everglades system is a special case where anthropogenic activities have affected its hydrology, nutrient status, and atmospheric deposition (Hurley et al., 1998). It is possible that DOC quality may restrict binding or that other ligands control Hg in some circumstances (Aiken et al., 2011; Hurley et al., 1998).

Although the vast majority of relationships (all Hg and DOC forms confounded) were positive, we unexpectedly found a few cases where relationships were negative. Negative relationships were not only found in studies conducted in systems impacted by anthropogenic activities such as urban areas (Brigham et al., 2009), intense industrialization (Fagnani et al., 2012), waste water (Guedron et al., 2014), artificial reservoirs (Noh et al., 2016), mining activities (Eckley et al., 2015), and near an old lithium industry (Riscassi et al., 2016) but also in natural systems (Bravo et al., 2017; Brazeau et al., 2013; Porvari & Verta, 2003; Vidon et al., 2014). A possible explanation for these negative relationships is that anthropogenic or natural inputs of Hg that are not accompanied by DOC inputs alter the basic relationship. This is consistent with the generally lower slopes that we found for systems subjected to human disturbance (Figure 2). In contaminated systems, Hg is likely bound to suspended matter (Brigham et al., 2009; Grigal, 2002) or to sulfide species such as cinnabar (Barnett et al., 1995; Miller et al., 2013) rather than to DOC which could explain the Hg–DOC decoupling. Riscassi and Scanlon (2011) found a negative relationship between Hg_D :DOC and total (wet and dry) Hg deposition using five sites from Brigham et al. (2009). On the other hand, Stoken et al. (2016) found no relationship between ratio of Hg_D :DOC and atmospheric wet Hg deposition across the US. These authors conclude that the amount of Hg freshly deposited is not the dominant factor explaining the spatial variability in the coupling between Hg and DOC and subsequent transport in the system (Riscassi & Scanlon, 2011; Stoken et al., 2016). In addition, the recent deposition may not accurately reflect the historical Hg accretion in soil due to a time lag between the two sources, which could explain the negative (or lack of) relationship between Hg :DOC and fresh Hg input (Stoken et al., 2016). Overall, negative relationships are rare in most systems and result from a decoupling between the sources and processes.

All studies in this meta-analysis reported linear Hg–DOC relationships without data transformation. A logarithmic relationship would indicate that the pool of DOC in the system is in excess and that virtually all the Hg is bound. On the other hand, exponential relationships would indicate that DOC is limiting the transport of Hg and that there is saturation of binding sites. The latter is not expected according to the literature (Stoken et al., 2016) but could occur in cases of high Hg deposition or near point source. Since most studies reported almost exclusively linear relationships, this suggests that environmental conditions are not limiting binding. However, a recent study by Stoken et al. (2016) regressed Hg :DOC ratios against the watershed SOC in several studies across the USA and found a negative power-law relationship. The authors referred to the concept of dilution by OM (Meili, 1991), which states that Hg availability in the system is the limiting factor in this situation when the carbon pool is in excess (Stoken et al., 2016). Those concepts are not entirely comparable (i.e., the linear Hg–DOC relationships in this meta-analysis and the power-law Hg :DOC–SOC relationships in Stoken et al., 2016); however, it suggests that at a regional scale, the amount of Hg is transported

proportionally to the amount of DOC available for adsorption, but that at a larger scale, the amount of Hg per unit of DOC is limited when carbon pools are in excess.

Variables explaining differences in strength in this study were continent, human disturbance, sampling design, and filtration, whereas latitude and longitude, ecosystem type (lentic vs. lotic), and publication year were not significant for both THg and MeHg. Predictors describing the type of human disturbance had the strongest effect on the relationship with DOC for both total and MeHg (Figure 2). Weaker relationships are expected when Hg and DOC sources are distinct from each other (Ravichandran, 2004; Riscassi & Scanlon, 2011; Stoken et al., 2016), which was the case for studies on systems affected by anthropogenic activities. Only continent was a significant factor, while coordinates were not. Strongest relationships were generally found in North America for THg and North America and Europe for MeHg. We hypothesize that differences between continents may be explained by differences in geomorphology, soil and water physico-chemical properties, and type of OM in the respective ecoprovinces, but we could not test this hypothesis with the available data. Hence, further studies will be needed to elucidate the causes of this pattern.

Spatial and spatiotemporal studies showed stronger relationships than temporal studies within a given system. Seasonal differences have been found to affect the coupling (Burns et al., 2013; Hurley et al., 1995), and this may partly explain weaker relationships for temporal studies. MeHg production is expected to decrease in colder months (Korthals & Winfrey, 1987; Ullrich et al., 2001), whereas DOC concentrations may not be noticeably lower during colder months (Wilson et al., 2013), which would reduce the temporal coupling with DOC. Filtration (dissolved vs. total) was a significant factor to determine the strength of the relationship. Hg contained in the dissolved fraction generally has high affinity with DOC (Babiarz et al., 2001; Ravichandran, 2004). However, in our case stronger relationships for dissolved Hg were only found for THg, whereas the dissolved and total MeHg had equally strong relationships with DOC. Several reviewed studies used filters with 0.45- μm pores, but many used 0.7- μm pores. Other studies used 0.2, 0.3, and 0.4- μm pores. Filtration has been historically set to an arbitrary cutoff of 0.45- μm pores, but the division is operational (Aiken et al., 2011). This cutoff removes suspended particles and phytoplankton from water but retains most of the colloidal fraction (Aiken et al., 2011; Mason, 2013). Moreover, the fraction that passes through those filters may contain a high proportion of Hg (20%–80%) that is bound to colloids (Babiarz et al., 2001; Hsu-Kim et al., 2013) and the word “dissolved” is therefore inaccurate. Overall, our results suggest that the coupling between Hg and DOC is stronger spatially (among sites) rather than temporally (within sites) and that discrepancies in filtration methods could explain some of the variation in the strengths of relationships.

We found generally weaker relationships (r) and shallower slopes for MeHg–DOC compared to THg–DOC relationships, as expected based on current empirical evidence (Burns et al., 2012; Hall et al., 2008; Lescord et al., 2018). The most likely explanation is that biological processes responsible for the production of 1 mole of MeHg do not necessarily result in a 1:1 change in DOC concentration, hence reducing the coupling with DOC. Moreover, the origin and molecular composition of OM are critical parameters to understand Hg methylation and accumulation in lakes and rivers (Bravo et al., 2017, 2018). Although Hall et al. (2008) found no significant relationship with DOC for MeHg, they did find significant relationships with HPOA and SUVA₂₅₄, confirming the importance of the type and quality of DOC. On one hand, high input of allochthonous DOC from the catchment may be less available to bacteria to promote methylation resulting in high concentrations but low methylation (Bravo et al., 2017). On the other hand, high in situ production of DOC (i.e., algal bloom) may decrease concentration but increase bacterial activity and in turn methylation (Bravo et al., 2017). In addition, photodemethylation of MeHg can increase with DOC concentration via photoproduction of reactive species (Girard et al., 2016; Hammerschmidt & Fitzgerald, 2010), which could weaken the MeHg–DOC relationship. Overall, qualitative aspects of the Hg and DOC cycles related to local biological reprocessing of both elements in freshwater ecosystems may lead to a lower coupling with MeHg compared to the stronger relationships that are typically observed for THg.

Along the same line, we found a statistically stronger (higher coefficient of correlation; r) relationships when the aromatic fraction of DOC (e.g., SUVA₂₅₄) was used as opposed to the bulk DOC for both THg and MeHg. This stronger relationship is not surprising because Hg is known to readily bind to DOC of high aromaticity where reduced sulfur binding sites are abundant and are generally exceeding the pool of Hg in natural systems (Liem-Nguyen et al., 2017; Xia et al., 1999). Stronger relationships were also found for SUVA₂₅₄ or HPOA than for bulk DOC in Louisiana, USA (Hall et al., 2008), and in northeastern U.S. streams

(Dittman et al., 2009). Burns et al. (2013) also found stronger relationships with $SUVA_{254}$ compared with bulk DOC but not with HPOA concentration in the Adirondack River basin. These results suggest that the bulk DOC pool contains a variety of molecules from both aquatic and terrestrial origins (Kellerman et al., 2015) but that the aromatic fraction of DOC, which is often associated to DOC of terrestrial origin (Lapierre & del Giorgio, 2014; Spencer et al., 2009), may be a better proxy of total Hg concentrations because it traces the comobilization of both elements at the landscape level.

We found a global average for the slope of 0.25 (CI: 0.20–0.31) ng Hg/mg DOC and a median of 0.19 ng/mg for THg and 0.029 (0.014–0.044) ng/mg and a median of 0.012 ng/mg for MeHg (Table 1). The values for THg were similar to the previously reviewed mean slope of 0.2 ng/mg for six Hg–DOC relationships (Grigal, 2002) but lower than reviewed mean Hg:DOC ratios of 0.46 (± 0.26 standard deviation) ng/mg (Riscassi & Scanlon, 2011) and 0.42 ng/mg (Stoken et al., 2016) that have been documented in lotic systems in the USA and Sweden. Note however, that the slope of a regression between y and x are not directly comparable to average ratios of x/y unless the intercept is equal to zero (Prairie et al., 1995). Average of intercept in our study was different than zero for THg (mean: 0.59, CI: 0.22–0.97, median: 0.52) but not for MeHg (mean: -0.019 , CI: -0.200 – 0.162 , median: 0.013), so ratios are therefore comparable to slopes for MeHg but not THg.

5. Conclusions

Our results highlight a widespread but variable coupling between mercury and dissolved organic carbon in freshwater ecosystems globally based on 52 studies from 14 countries on four continents. The coupling varies across spatial and temporal scales and appears to be driven by geographic factors such as location and anthropogenic influence. Interestingly, relationships were stronger when including several different sites sampled once (spatial design) than on a single site with repeated sampling (temporal design). Filtration had an impact on the strength of THg relationships but made no difference for MeHg, which indicates both the importance of the dissolved fraction for the affinity between Hg and DOC and the diverging processes governing the two Hg species. On average, our results indicate that for every additional mg of DOC in the world's lakes, reservoirs and rivers, there tends to be an increase of 0.25 ng of Hg and 0.029 ng of MeHg because Hg and DOC share transport and transformation processes at the landscape level. The ecosystem type (lentic vs. lotic) was surprisingly not a significant variable to explain variations in the coupling. Geographical coordinates and publication year also had no effect on the strength of relationships. Relationships were stronger with aromatic DOC where reduced sulfur binding sites are abundant than with total DOC. This meta-analysis allowed to reach conclusions that would be difficult to achieve for a fieldwork study. For instance, the comparison of spatial and temporal scales as well as the comparison among continents would require an impressive and costly sampling design. The global scope of this study allowed us to identify geographic variables driving the direction of the Hg–DOC relationships. We propose that a better understanding of the molecular underpinnings of the DOC and Hg cycling in freshwaters may shed new light on Hg–DOC coupling. There is a growing interest in using DOC characterization techniques to improve our understanding of Hg dynamics in freshwaters, including $SUVA_{254}$ (and other wavelength), HPOA (Burns et al., 2013; Dittman et al., 2009), parallel factor analysis modeling of DOC fluorescence (Jiang et al., 2018; Lescord et al., 2018), semiquantitative analytical methods by a pyrolysis–gas chromatography–mass spectrometry (Py–GC–MS) (Bravo et al., 2017), and fully quantitative liquid chromatography (LC)–MS or GC–MS analyses. Such analyses may help better quantifying Hg affinity with DOC under varying environmental conditions. Characterization of mercury speciation, however, was not as routinely used in studies survey in this meta-analysis and this would be important to quantify. A greater understanding on how DOC and Hg speciation explains coupling and bioavailability would fill important gaps in knowledge at the chemical level, which may explain broader-scale patterns over time and space.

At the landscape level, we argue that future studies should focus on the effect of geographic drivers on the differential mobilization and transformation of Hg and DOC in soils and in freshwater ecosystems. For instance, Stoken et al. (2016) found no impact of Hg deposition on Hg:DOC ratios but found a negative effect of SOC concentrations due to a dilution effect. Along the same line, explicit consideration of geographic drivers such as runoff, slope, and precipitations that control the mobilization of DOC from soils to water and the exposure time to microbial and photochemical processes in the water should improve predictions and may resolve the cross-regional differences in the slopes and intercepts of the Hg–DOC relationships. We

suggest that filling these knowledge gaps will enhance our capacity to estimate concentrations of mercury in freshwaters based on the widely available organic carbon data, as well as our ability to anticipate the direction and magnitude of the changing mercury and carbon cycles in response to rapid changes in climate (including permafrost thaw) and land use worldwide.

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